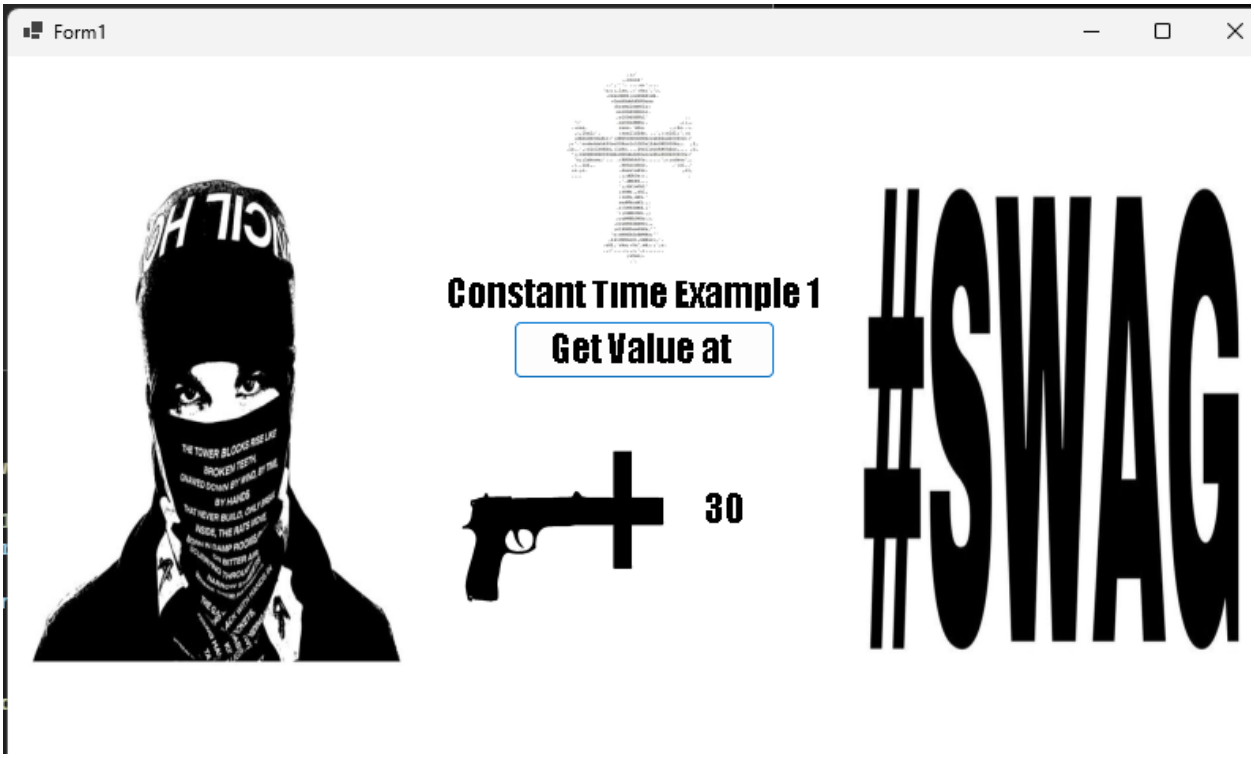
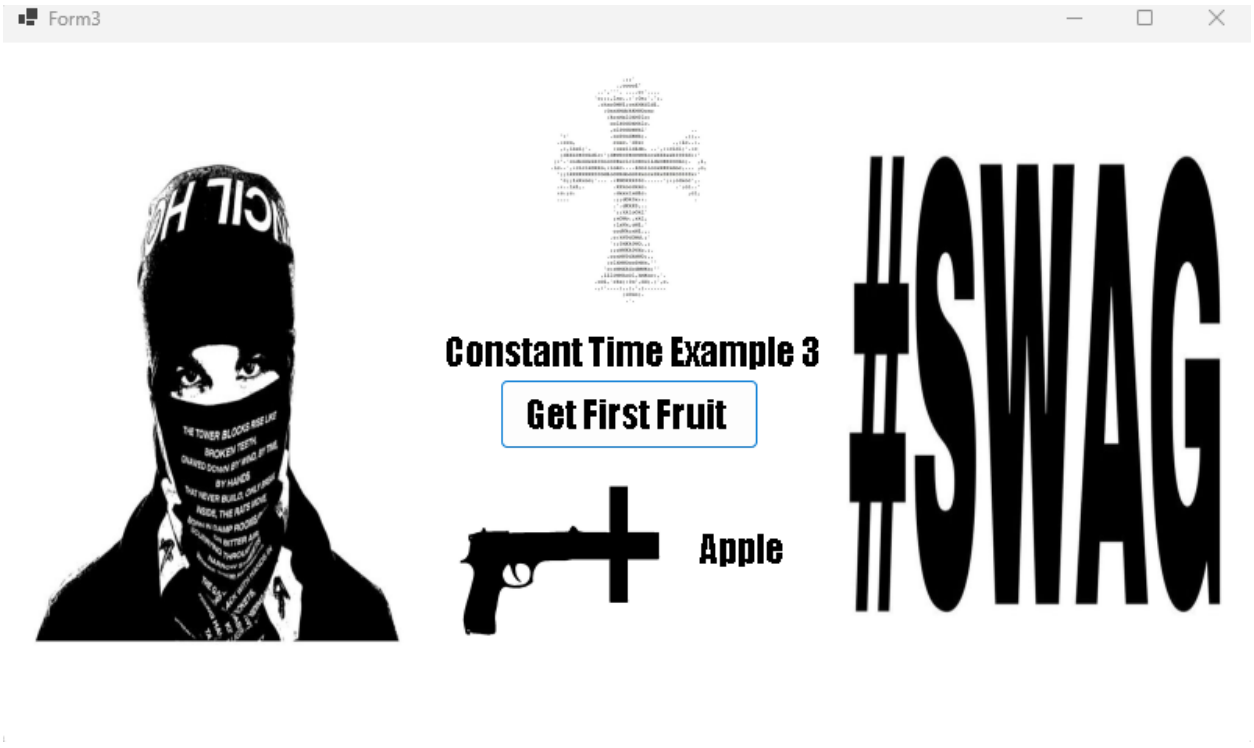



```
}

1 reference
private void btnGetAnswer_Click(object sender, EventArgs e)
{
    int[] numbers = { 10, 20, 30, 40, 50 };
    int answer = numbers[2];

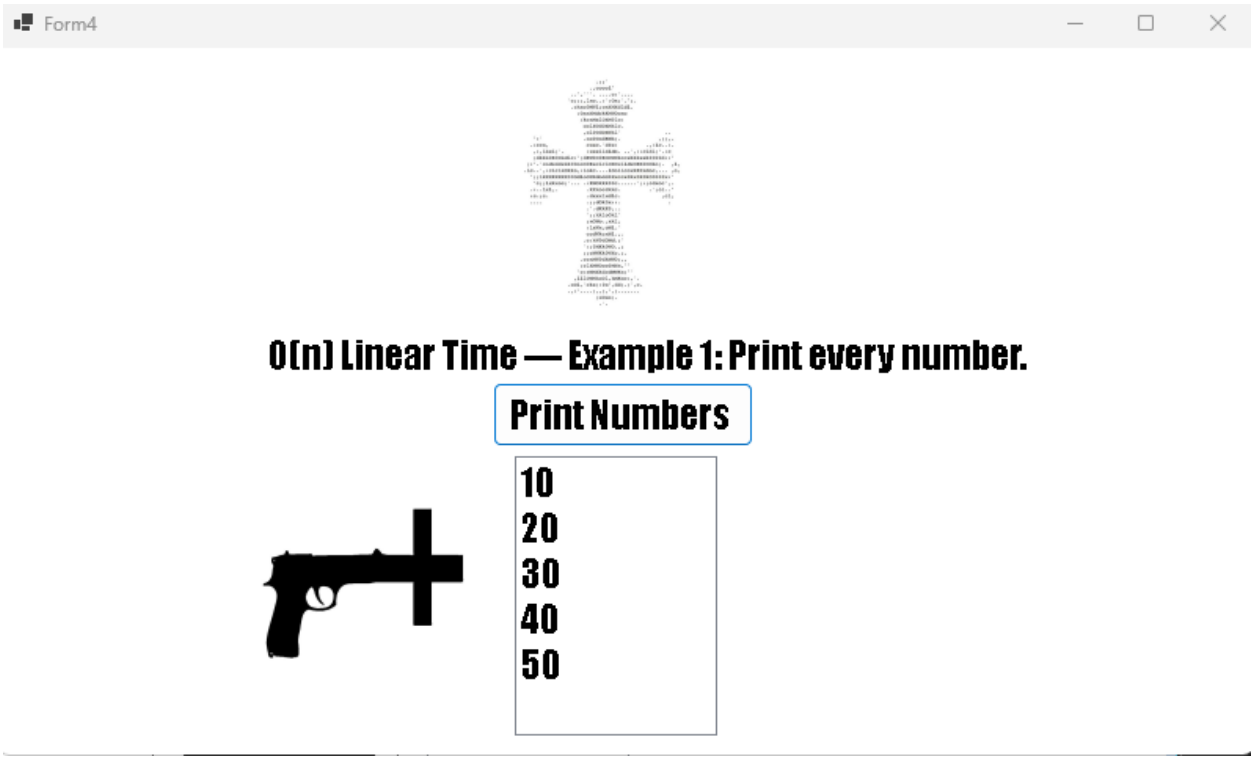
    lblAnswer.Text = answer.ToString();
}

1 reference
private void Form1_Load(object sender, EventArgs e)
{
```




```
private void btnGetFirstItem_Click(object sender, EventArgs e)
{
    string[] fruits = { "Apple", "Banana", "Mango" };
    string fruit = fruits[0];


    lblFruitResult.Text = fruit;
}
```

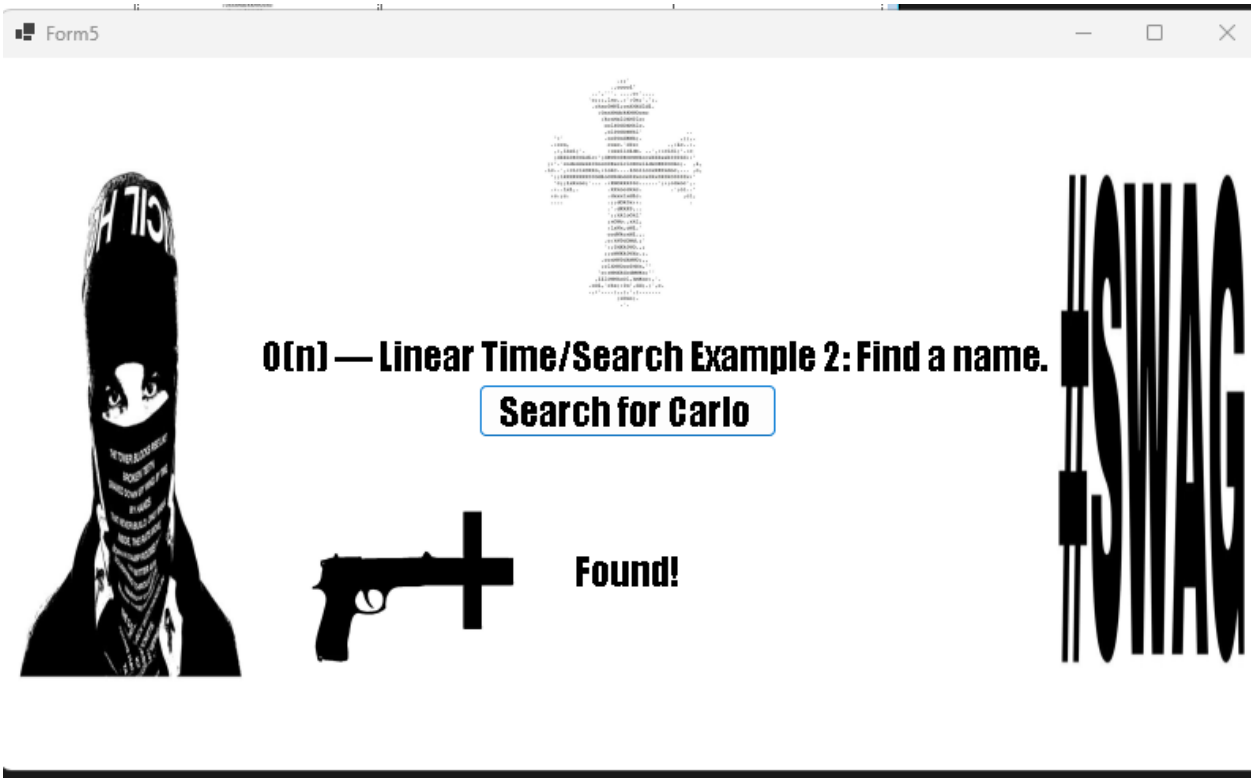



```
1 reference
private void btnPrintNumbers_Click(object sender, EventArgs e)
{
    lstNumbers.Items.Clear();


    int[] numbers = { 10, 20, 30, 40, 50 }; //

    for (int i = 0; i < numbers.Length; i++)  //[cite: 9]
    {

        lstNumbers.Items.Add(numbers[i]);
    }
```




```
1 reference
private void btnFindName_Click(object sender, EventArgs e)
{
    string[] names = { "Ana", "Ben", "Carlo", "David" };

    // Clear previous output
    lblSearchResult.Text = "";

    // Exact loop from slide
    foreach (string name in names)
    {
        if (name == "Carlo")
        {
            // Replaces Console.WriteLine("Found!");
            lblSearchResult.Text = "Found!";
        }
    }
```

Form6





O(n) — Linear Time/Search Example 3: Calculate the total.

Calculate Total



30




```
1 reference
private void btnCalculateTotal_Click(object sender, EventArgs e)
{
    int[] numbers = { 5, 10, 15 };
    int total = 0;

    foreach (int number in numbers)
    {
        total = total + number;
    }


    lblTotalResult.Text = total.ToString();
}
```

Form7





O(log n) — Logarithmic Time Example 1: Binary Search.

Search for 50



Found!




```
1 reference
private void btnBinarySearch_Click(object sender, EventArgs e)
{
    int[] numbers = { 10, 20, 30, 40, 50, 60, 70 };

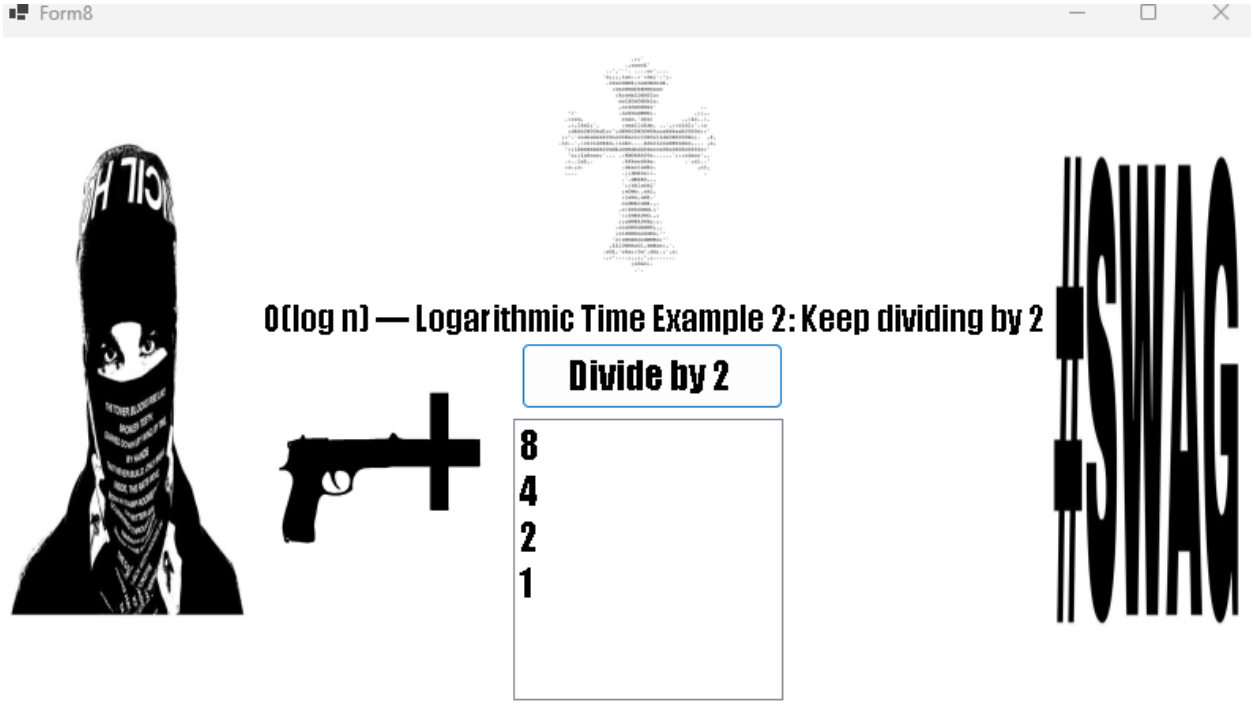
    int left = 0;
    int right = numbers.Length - 1;

    lblBinaryResult.Text = "";

    while (left <= right)
    {
        int middle = (left + right) / 2;

        if (numbers[middle] == 50)
        {

            lblBinaryResult.Text = "Found!";
            break;
        }
        else if (numbers[middle] < 50)
        {
            left = middle + 1;
        }
        else
        {
            right = middle - 1;
        }
```

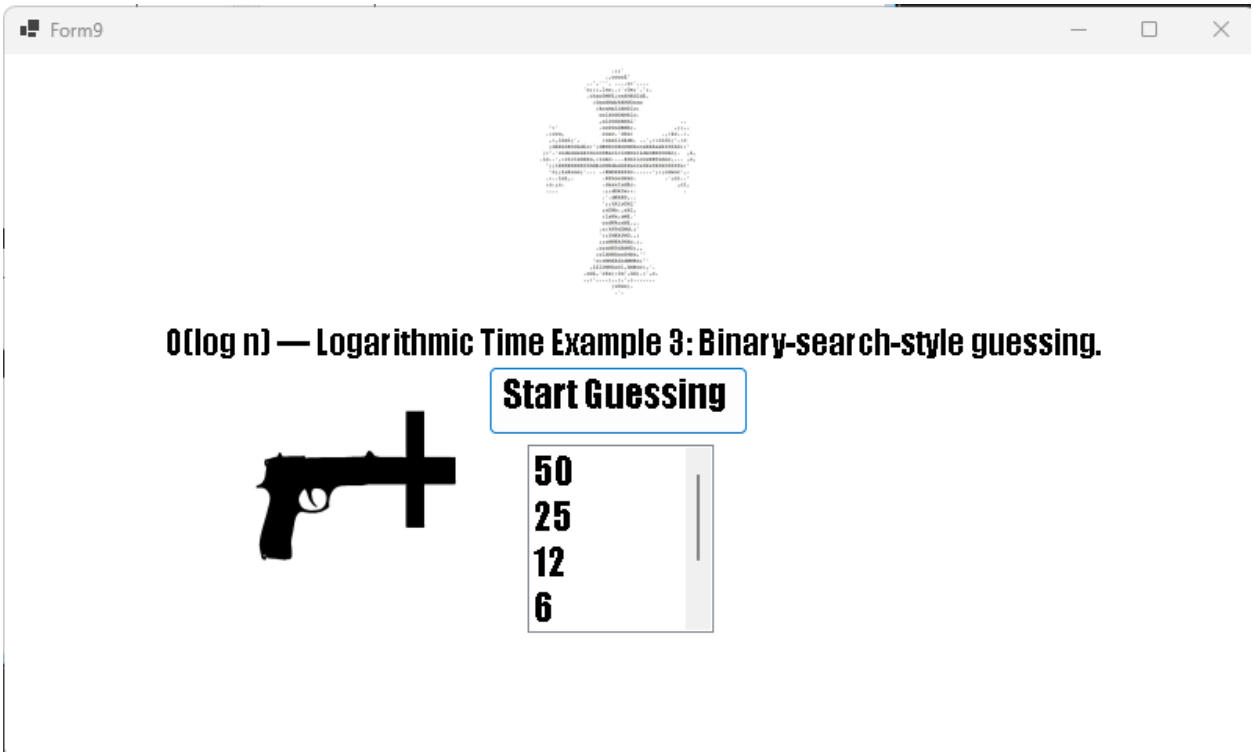



```
1 reference
private void btnDivide_Click(object sender, EventArgs e)
{
    lstDivideResults.Items.Clear();


    int n = 16;

    while (n > 1)
    {
        n = n / 2;


        lstDivideResults.Items.Add(n);
    }
}
```



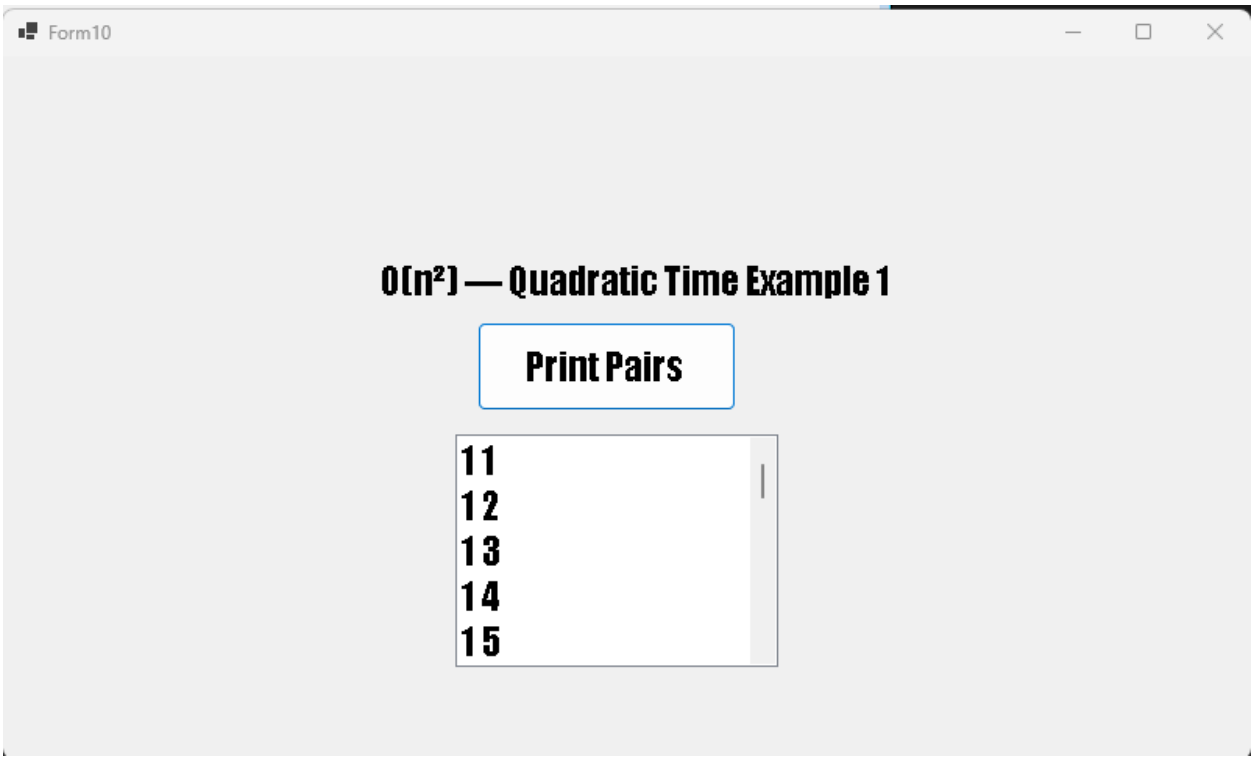

```
1 reference
private void btnGuess_Click(object sender, EventArgs e)
{
    lstGuesses.Items.Clear();


    int low = 1;
    int high = 100;

    while (low <= high)
    {
        int middle = (low + high) / 2;


        lstGuesses.Items.Add(middle);

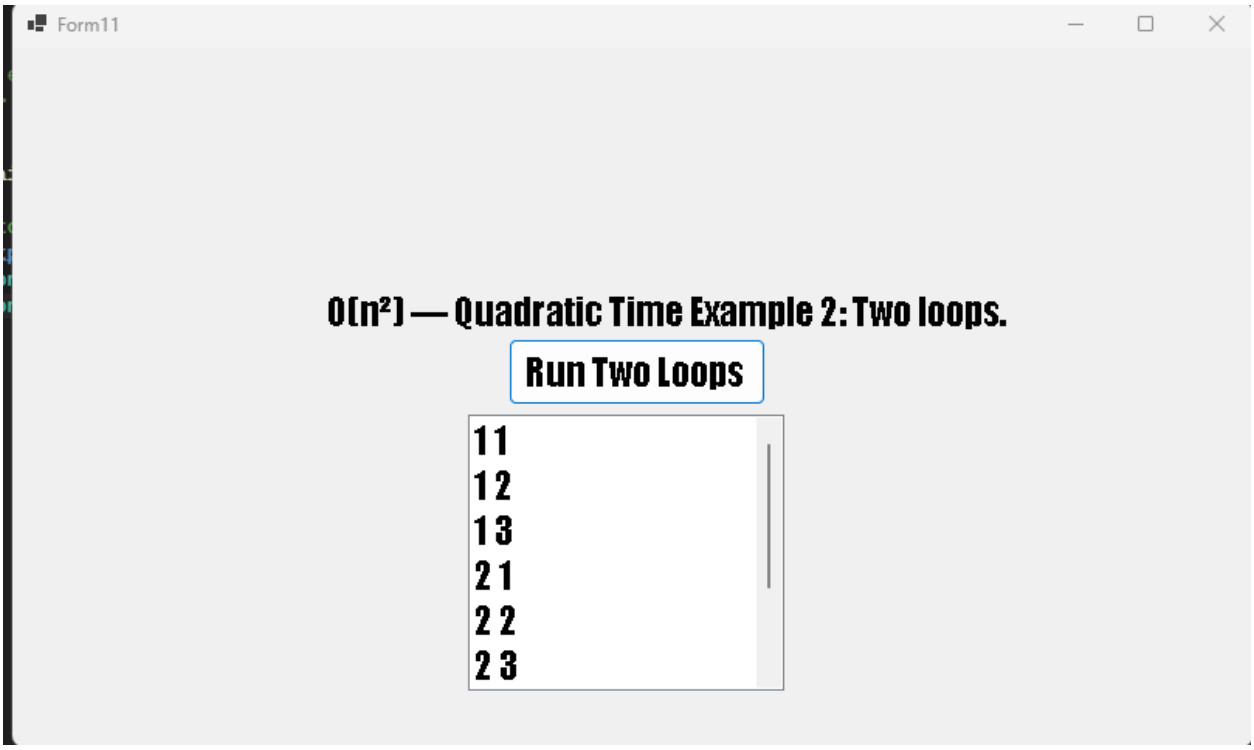
        high = middle - 1;
    }
```




```
1 reference
private void btnPrintPairs_Click(object sender, EventArgs e)
{
    lstPairs.Items.Clear();


    int[] numbers = { 1, 2, 3, 4, 5 };

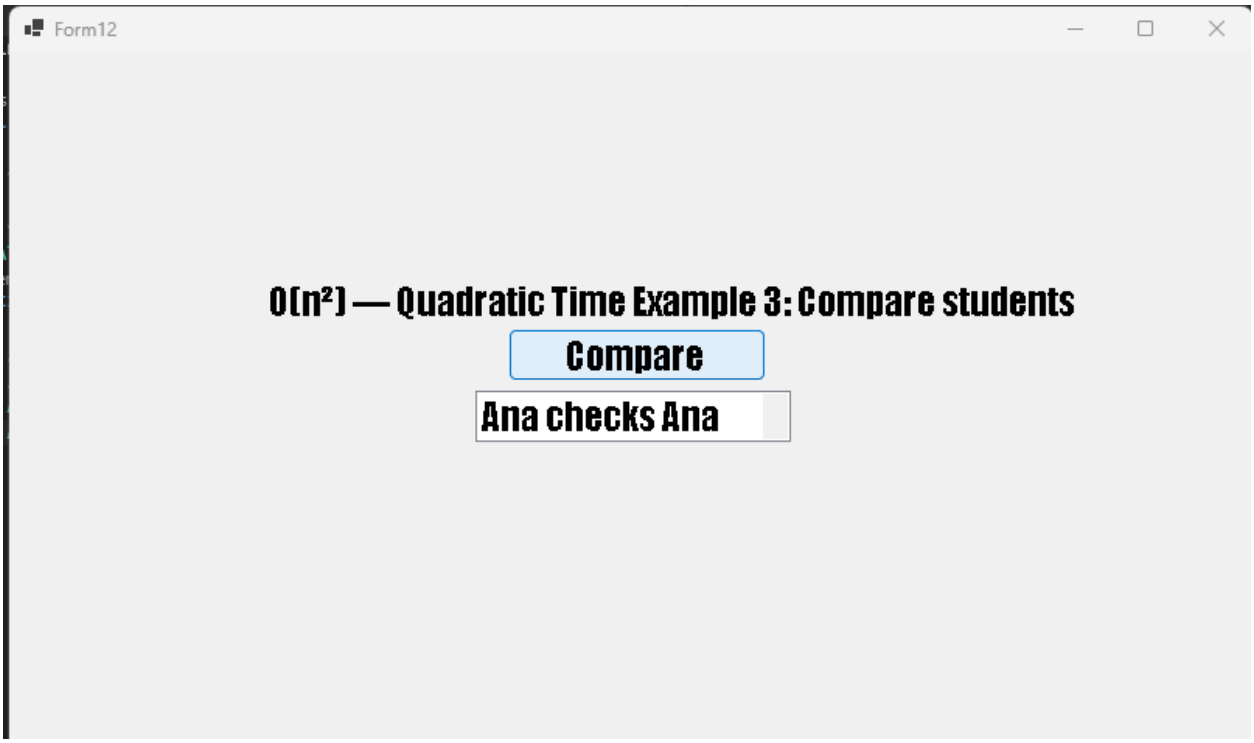
    for (int i = 0; i < numbers.Length; i++)
    {
        for (int j = 0; j < numbers.Length; j++)
        {
            // Replaces Console.WriteLine(numbers[i] + " " + numbers[j]);
            lstPairs.Items.Add(numbers[i] + " " + numbers[j]);
        }
    }
}
```




```
1 reference
private void btnTwoLoops_Click(object sender, EventArgs e)
{
    lstTwoLoopsResults.Items.Clear();


    int[] numbers = { 1, 2, 3 };

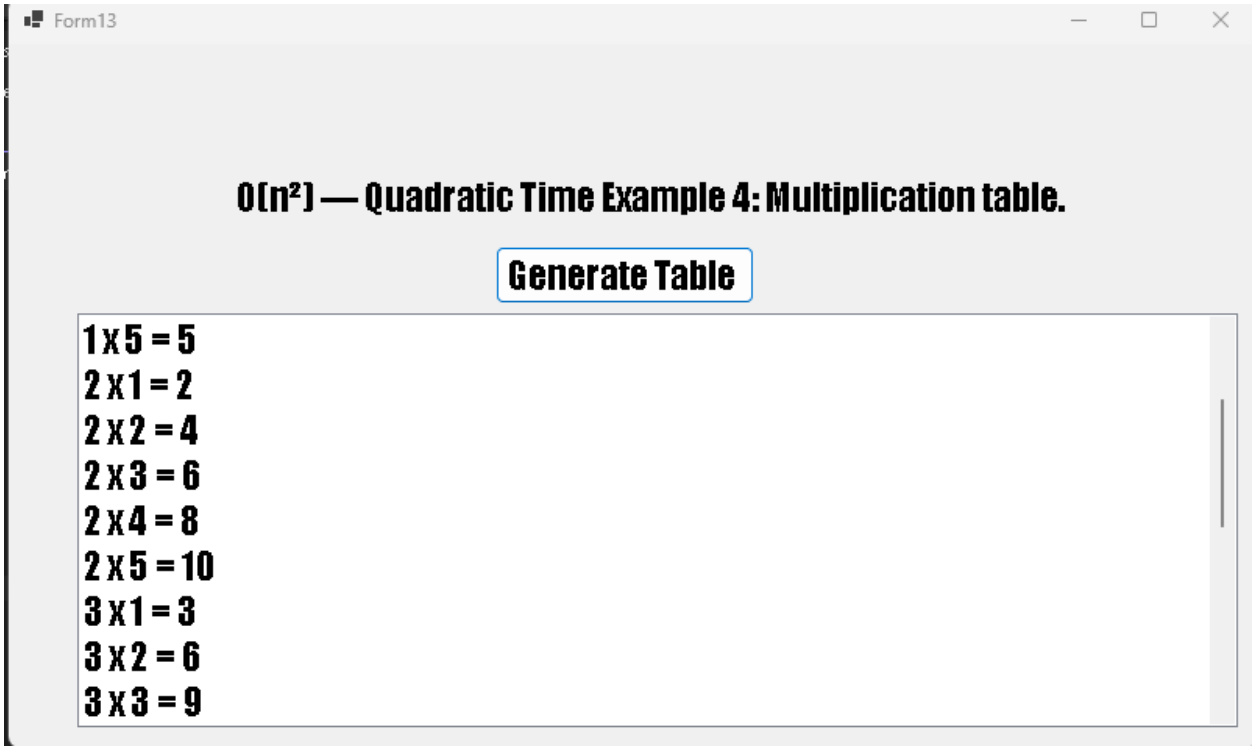
    for (int i = 0; i < numbers.Length; i++)
    {
        for (int j = 0; j < numbers.Length; j++)
        {
            // Replaces Console.WriteLine(numbers[i] + " " + numbers[j]);
            lstTwoLoopsResults.Items.Add(numbers[i] + " " + numbers[j]);
        }
    }
}
}
```




```
1 reference
private void btnCompareStudents_Click(object sender, EventArgs e)
{
    lstCompareResults.Items.Clear();


    string[] students = { "Ana", "Ben", "Carlo" };

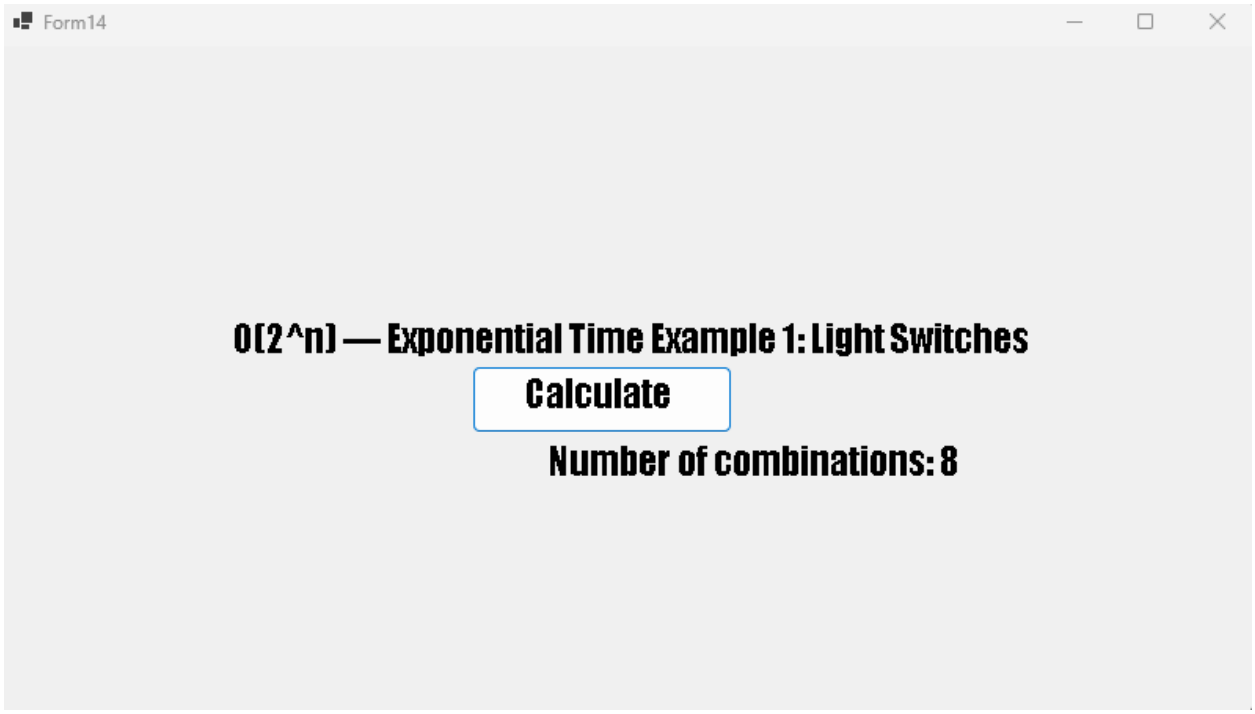
    for (int i = 0; i < students.Length; i++)
    {
        for (int j = 0; j < students.Length; j++)
        {
            // Replaces Console.WriteLine(students[i] + " checks " + students[j]);
            lstCompareResults.Items.Add(students[i] + " checks " + students[j]);
        }
    }
}
```




```
1 reference
private void btnMultiplicationTable_Click(object sender, EventArgs e)
{
    lstMultiplicationResults.Items.Clear();


    for (int i = 1; i <= 5; i++)
    {
        for (int j = 1; j <= 5; j++)
        {

            lstMultiplicationResults.Items.Add(i + " x " + j + " = " + (i * j));
        }
    }
}
}
```

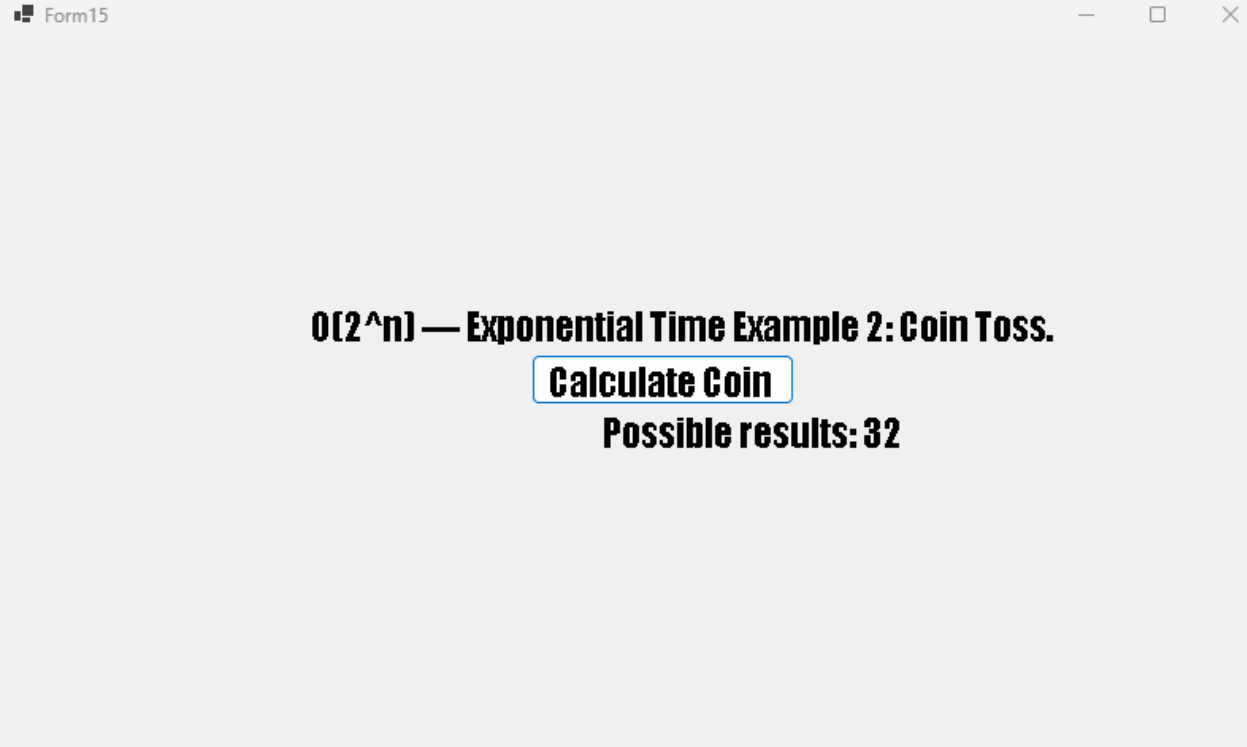



```
1 reference
private void btnCalculateCombinations_Click(object sender, EventArgs e)
{
    int n = 3;
    int combinations = 1;

    for (int i = 0; i < n; i++)
    {
        combinations = combinations * 2;

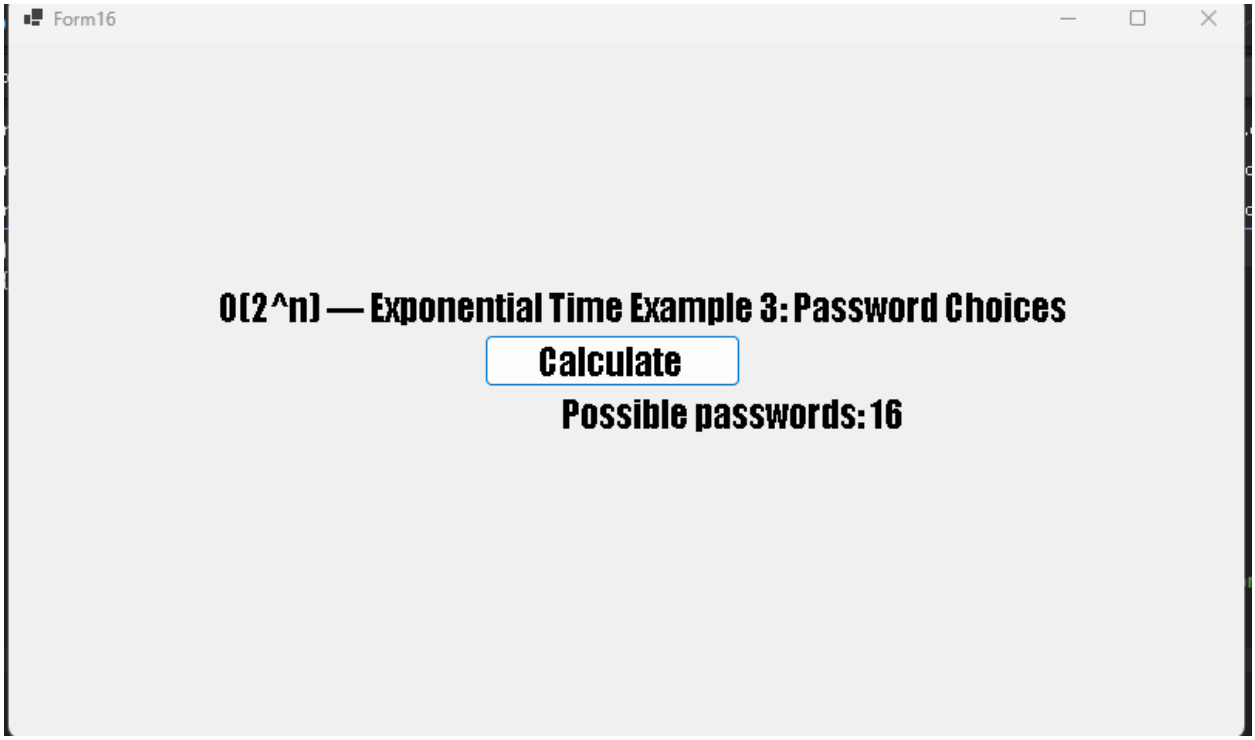
        lblCombinationsResult.Text = "Number of combinations: " + combinations;
    }
}

1 reference
```




```
1 reference
private void btnCoinToss_Click(object sender, EventArgs e)
{
    int tosses = 5;
    int possibilities = 1;

    for (int i = 0; i < tosses; i++)
    {
        possibilities *= 2;
    }
```




```
        }

        1 reference
        private void btnPasswordChoices_Click(object sender, EventArgs e)
        {
            int positions = 4;
            int possibilities = 1;

            for (int i = 0; i < positions; i++)
            {
                possibilities *= 2;
            }


            lblPasswordResult.Text = "Possible passwords: " + possibilities;

    }
    }
}
```